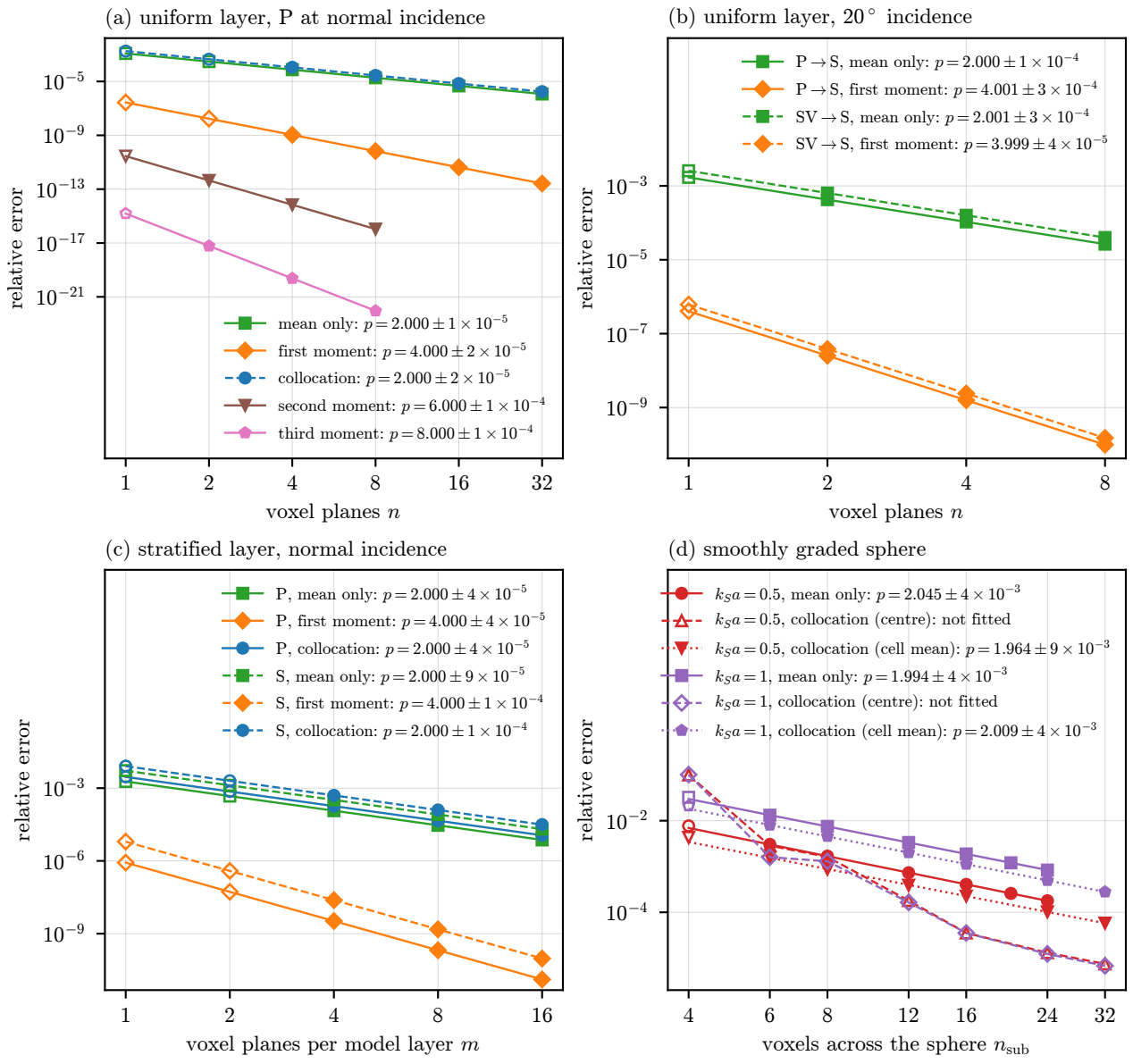
(a) uniform layer, P at normal incidence
relative error
10^{-5}
10^{-9}
10^{-13}
10^{-17}
10^{-21}
mean only: $p = 2.000 \pm 1 \times 10^{-5}$
first moment: $p = 4.000 \pm 2 \times 10^{-5}$
collocation: $p = 2.000 \pm 2 \times 10^{-5}$
second moment: $p = 6.000 \pm 1 \times 10^{-4}$
third moment: $p = 8.000 \pm 1 \times 10^{-4}$
1
2
4
8
16
32
voxel planes n
(b) uniform layer, 20° incidence
P → S, mean only: $p = 2.000 \pm 1 \times 10^{-4}$
P → S, first moment: $p = 4.001 \pm 3 \times 10^{-4}$
SV → S, mean only: $p = 2.001 \pm 3 \times 10^{-4}$
SV → S, first moment: $p = 3.999 \pm 4 \times 10^{-5}$
relative error
10^{-3}
10^{-5}
10^{-7}
10^{-9}
1
2
4
8
voxel planes n
(c) stratified layer, normal incidence
P, mean only: $p = 2.000 \pm 4 \times 10^{-5}$
P, first moment: $p = 4.000 \pm 4 \times 10^{-5}$
P, collocation: $p = 2.000 \pm 4 \times 10^{-5}$
S, mean only: $p = 2.000 \pm 9 \times 10^{-5}$
S, first moment: $p = 4.000 \pm 1 \times 10^{-4}$
S, collocation: $p = 2.000 \pm 1 \times 10^{-4}$
relative error
10^{-3}
10^{-6}
10^{-9}
1
2
4
8
16
voxel planes per model layer m
(d) smoothly graded sphere
$k_S a = 0.5$, mean only: $p = 2.045 \pm 4 \times 10^{-3}$
$k_S a = 0.5$, collocation (centre): not fitted
$k_S a = 0.5$, collocation (cell mean): $p = 1.964 \pm 9 \times 10^{-3}$
$k_S a = 1$, mean only: $p = 1.994 \pm 4 \times 10^{-3}$
$k_S a = 1$, collocation (centre): not fitted
$k_S a = 1$, collocation (cell mean): $p = 2.009 \pm 4 \times 10^{-3}$
relative error
10^{-2}
10^{-4}
4
6
8
12
16
24
32
voxels across the sphere n_{sub}